A Recursive-Additive Closure for Chain-Weighted Superlative Growth Rates

Jaewon Park[1*]

[1]Department of Economics, George Mason University, 4400 University Dr, Fairfax, 22030, VA, U.S.A.

*Corresponding Author E-mail(s): jpark261@gmu.edu;

Abstract

Chain-weighted superlative indexes deliver economically meaningful growth rates but do not generate real levels that add up across components, creating persistent accounting inconsistencies in applied work. We argue that this non-additivity reflects a missing intertemporal accounting closure rather than a limitation of superlative index theory. We introduce a recursive-additive aggregation operator that takes official component superlative growth rates as primitives and constructs an exactly additive system of real levels over time. Exact additivity is enforced by defining the aggregate as the sum of recursively accumulated components, while aggregate growth emerges endogenously. The resulting growth deviation from the published chain aggregate is of higher order in value-share variation. Using BEA industry data, we document that conventional chain-weighted levels exhibit large and persistent non-additivity, whereas the growth distortion induced by recursive additivity is negligible and robust to base-year normalization.

Keywords: chain-weighted indexes, exact additivity, recursive aggregation, growth accounting

JEL Classification: C43 , E01 , O47 , E23

1 Introduction

A central objective of modern national accounts is to measure real economic activity in a manner that is both economically meaningful and internally coherent. Since the widespread adoption of chain-weighted superlative index numbers, official statistics have largely succeeded in the first dimension. Fisher, Törnqvist, and related superlative formulas deliver growth rates that closely approximate changes in an underlying aggregator and substantially mitigate substitution bias relative to fixed-base measures (Diewert 1978; Caves et al. 1982; Whelan 2000). As a result, reported growth rates of output, inputs, and productivity are widely regarded as economically interpretable.

At the same time, these same measures systematically fail in the second dimension. Published chain-weighted real levels are generally not exactly additive across components. Outside the reference normalization, aggregate real output does not equal the sum of sectoral real outputs by construction. Consequently, basic accounting identities that are routinely required in applied work—such as cumulative growth accounting, structural decomposition, and constant-price consistency—fail mechanically, even when growth rates themselves are well defined.

This non-additivity is neither new nor accidental. It has been carefully documented in both official statistics and the index-number literature. Statistical agencies explicitly caution users against summing chained-dollar components, treating non-additivity as an unavoidable cost of preserving economically meaningful growth rates (Reinsdorf et al. 2002; Landefeld et al. 2003; Whelan 2000). From a theoretical perspective, the failure of exact adding-up is understood as a consequence of time-varying weights: when relative prices change, deflation and aggregation do not generally commute, so no single static index-number formula can guarantee exact additivity in levels while retaining superlative properties (Pollak 1975; Balk 2004; Balk and Reich 2008). The prevailing conclusion is therefore that exact additivity and superlative economic meaning are fundamentally incompatible within the standard static framework.

This paper argues that this conclusion is too strong and, more importantly, misdirected. The core difficulty does not lie in superlative index theory itself, but in the way superlative growth rates are accumulated into multi-period level systems. Superlative theory is inherently local and two-period: it justifies adjacent-period growth comparisons but does not uniquely specify an intertemporal accounting rule that maps those growth primitives into a globally coherent system of real levels. Existing additive decompositions—including contribution-to-growth frameworks—restore adding-up only locally for growth contributions and do not close the accounting system in levels over time (Reinsdorf et al. 2002; Balk 2020). As a result, non-additivity arises not as a theoretical contradiction, but as a symptom of a missing intertemporal accounting closure.

The central contribution of this paper is to make that missing closure explicit. We propose a recursive-additive (RA) accounting operator that takes official component-level superlative growth rates as primitives and constructs an exactly additive system of real levels over time. The operator accumulates component levels recursively and defines the aggregate at each date as the sum of components, thereby enforcing exact additivity in levels by construction. Aggregate growth is treated as an implied object of the closed system rather than as an additional primitive. Under standard regularity conditions, the implied aggregate growth rate differs from the published chain aggregate only by higher-order terms in value-share variation, preserving local superlative fidelity while restoring global accounting coherence.

We illustrate the framework using U.S. industry value-added data. The empirical results show that non-additivity in published chained levels is sizable, episodic, and economically meaningful, while the growth-rate distortion induced by enforcing exact additivity is tightly concentrated near zero and robust to base-year normalization. These findings imply that the practical cost of restoring exact additivity is quantitatively small relative to the level inconsistencies embedded in conventional chained systems, and that the long-standing additivity problem reflects an accounting choice rather than an unavoidable measurement trade-off.

2 Positioning in the Index-Number and National-Accounts Literature

This paper sits at the intersection of (i) superlative index-number theory, (ii) chain-weighted national-accounts practice, and (iii) applied growth accounting that requires exact adding-up across components. The first pillar establishes why statistical agencies and productivity researchers rely on superlative formulas: Fisher, Törnqvist, and related indices deliver second-order accurate measures of economic change by approximating an unknown aggregator locally, via quadratic-approximation arguments and flexible functional forms (Diewert 1978, 2002; Caves et al. 1982). This foundational result is inherently two-period: it justifies adjacent-period growth comparisons, not a globally unique level system over long horizons. In parallel, the Divisia tradition clarifies why superlative indices are best interpreted as growth-rate objects and why their economic meaning is fundamentally local (Hulten 1973; Vartia 1976). Related contributions in monetary aggregation and welfare measurement reinforce the same message: superlative and differential indices provide economically interpretable changes, but they do not by themselves supply intertemporal accounting closure for levels (Barnett 1980; Barnett et al. 2003; Allen and Diewert 1981).

The second pillar documents the practical consequence of adopting such indices in official statistics. Chain weighting updates weights as relative prices move, mitigating substitution bias relative to fixed-base measures, and thereby improves the interpretability of reported growth rates (Whelan 2000; Landefeld et al. 2003). The same design choice, however, breaks exact additivity of chained-volume levels across components. This non-additivity is not a computational nuisance; it follows from classic aggregation results showing that no static index-number formula can generally satisfy superlative properties, time-reversal-type desiderata, and exact adding-up simultaneously when weights vary (Pollak 1975; Blair and Pollak 1982; Balk 2004; Balk and Reich 2008). Consequently, statistical agencies emphasize that chained dollars are not designed for component summation, and applied users are steered toward contribution-to-growth style diagnostics rather than additive level accounting.

A large applied literature responds by constructing additive decompositions of superlative growth—most prominently the contribution-to-growth framework and closely related linearizations. These approaches deliver additivity of growth contributions within a period or around a benchmark, while retaining the official superlative growth concept (Reinsdorf et al. 2002; Dumagan 2011a,b). Yet these decompositions are typically local: they do not define a closed intertemporal operator that maps a full history of component growth rates into an exactly additive sequence of real levels. As emphasized in the index-number decompositions literature, once one moves beyond a single comparison (or a single benchmark), exact additivity of levels is generally lost unless an additional accounting closure is imposed (Balk 2020; Balk and Zofío 2018).

This paper's contribution is to make that missing closure explicit. Rather than proposing a new index-number formula, we take component superlative growth rates as primitives and define a recursive accounting operator that produces an exactly additive system of real levels at every date. Conceptually, the proposed operator separates two objects that are often conflated in practice: (i) the locally meaningful superlative growth-rate statistic and (ii) the global intertemporal mapping from growth rates into additive levels. By treating the latter as an accounting choice—and by providing a closed-form recursive closure—we reconcile the needs of growth accounting (exact adding-up in levels) with the economic fidelity of superlative measurement (local second-order accuracy), thereby reframing chained non-additivity as structural incompleteness of the level-accumulation convention rather than a limitation of superlative theory itself.

3 The Accounting Problem and Framework

3.1 The Mapping Problem: From Growth Rates to Additive Levels

Modern national accounts and productivity statistics rely on chain-weighted superlative index numbers to deliver economically meaningful measures of growth. These growth rates are treated as primitives: they are constructed by statistical agencies and are taken as given by applied researchers. The central problem addressed in this paper does not concern the choice or estimation of these growth rates. Instead, it concerns how such growth rates are accumulated into real levels that satisfy fundamental accounting identities.

Formally, suppose that for each component $i = 1, \ldots, N$ and for the aggregate, a sequence of superlative log-growth rates is observed. These growth rates are internally consistent in the sense of index number theory and are designed to approximate changes in an underlying aggregator function (Diewert 1978; Caves et al. 1982). The unresolved question is how to construct a corresponding system of real levels such that, at every point in time, the aggregate real level equals the sum of component real levels.

This problem can be stated precisely. Let $\{g_{i,t}\}_{i=1}^{N}$ denote component log-growth rates and let g_t denote the published aggregate log-growth rate. A level system is a sequence $\{Q_{i,t}\}_{t\geq 0}$ that generates the observed growth primitives:

$$\Delta \log Q_{i,t} = g_{i,t}, \qquad i = 1, \ldots, N,\ t \geq 1. \tag{1}$$

Chaining each component independently pins down $\{Q_{i,t}\}$ only up to initial conditions, but it does not guarantee that there exists any aggregate sequence $\{Q_t\}$ such that

$$Q_t = \sum_{i=1}^{N} Q_{i,t} \quad \text{for all } t \geq 0, \qquad \Delta \log Q_t = g_t \quad \text{for all } t \geq 1. \tag{2}$$

Equation (2) makes the mapping problem explicit: the published growth primitives do not uniquely—and in general do not jointly—determine a globally additive level system. Without additional intertemporal structure, the accumulation of superlative growth rates into levels is underdetermined. This indeterminacy manifests empirically as the well-known non-additivity of chain-weighted real aggregates (Whelan 2000; Reinsdorf et al. 2002).

The difficulty arises because superlative index numbers are fundamentally local objects. They are designed to track economically meaningful changes between adjacent observations, not to define a globally consistent level system over time (Balk 2004; Diewert 2002). As a result, exact additivity across components is not guaranteed once growth rates are accumulated over multiple periods.

3.2 Why Existing Approaches Cannot Resolve the Problem

Several strands of the literature address aspects of aggregation and additivity, but none provides a complete solution to the mapping problem defined above.

First, static index number formulas are inherently two-period objects. While they deliver growth rates with strong economic interpretation, they do not specify how these rates should be accumulated into levels across multiple periods. Chaining links successive growth rates mechanically, but chaining alone does not impose exact additivity on the resulting levels (Pollak 1975; Balk and Reich 2008).

Second, additive decompositions of superlative indexes achieve exact additivity only locally. Methods such as the BEA contribution-to-growth framework construct additive component contributions within a single comparison or around a benchmark year (Reinsdorf et al. 2002; Dumagan 2011a). While these decompositions are highly informative for descriptive analysis, they do not define a closed intertemporal accounting system. Once contributions are accumulated over time, exact additivity generally fails (Balk 2020).

Third, rebasing and normalization conventions are often used in practice to mitigate non-additivity. However, rebasing merely rescales existing level series and does not supply a missing accumulation rule. Different rebasing choices can therefore lead to different implied level paths, highlighting the absence of a uniquely defined additive system consistent with the observed growth rates (Whelan 2000).

Viewed through the requirements below, these limitations are structural. Two-period superlative formulas secure economically meaningful growth rates but are silent about a multi-period level law. Additive decompositions impose adding-up only locally and lack intertemporal closure. Rebasing alters normalization but cannot restore a missing accounting operator. In short, the measurement problem is not the construction of superlative growth rates; it is the absence of a globally defined accumulation rule that closes the accounting system in levels.

3.3 Requirements of a Valid Intertemporal Accounting System

Resolving the mapping problem in (2) requires more than a choice of index-number formula. It requires an explicit intertemporal accounting system that translates observed growth-rate primitives into real levels in a manner that is both economically interpretable and internally coherent. The demands placed on real aggregates in applied work implicitly impose a set of minimal requirements on any such system. We make these requirements explicit.

R1. Exact Additivity

At every date, the aggregate real level must equal the sum of component real levels:

$$Q_t = \sum_{i=1}^{N} Q_{i,t}.$$

Exact additivity is not a theoretical luxury. It is a basic accounting identity required for multi-period growth accounting, sectoral decomposition, and constant-price consistency in applications such as input–output analysis and productivity accounting (Hulten 1973; Parker and Triplett 1996). Without this identity, cumulative contributions and level-based comparisons across components become inherently ambiguous.

R2. Fidelity to Published Growth Primitives

The system must preserve the published component growth primitives exactly, i.e.,

$$\Delta \log Q_{i,t} = g_{i,t} \quad \text{for all } i, t.$$

These growth rates are the primary objects delivered by statistical agencies and are grounded in superlative index-number theory. A valid accounting framework must therefore treat them as primitives rather than redefining or approximating them. For the aggregate, fidelity should be understood in the superlative sense: any discrepancy between constructed and published aggregate growth should be theoretically higher order and quantitatively small under standard regularity conditions.

R3. Intertemporal Closure and Consistency

The accounting system must specify an explicit intertemporal accumulation rule mapping growth primitives into levels over the entire sample path. Levels must evolve recursively, such that today's level depends only on yesterday's level and the intervening growth rate. Without such closure, the accumulation of growth rates into levels is underdetermined, leaving key accounting identities contingent on ad hoc conventions rather than on a well-defined law of motion.

R4. Normalization Invariance

The resulting level system should be invariant to arbitrary normalization choices, such as the selection of a base year or rebasing convention. Different normalizations should yield proportionally equivalent representations of the same underlying real quantities, ensuring that economically meaningful objects—growth rates, shares, and relative comparisons—are not artifacts of an arbitrary benchmark.

These requirements clarify why existing approaches fall short. Superlative index numbers secure growth-rate fidelity (R2) but do not, by themselves, define an intertemporal level system satisfying exact additivity (R1) and closure (R3). Additive decompositions restore adding-up only locally for growth contributions and therefore fail to deliver intertemporal consistency in levels. Mechanical chaining supplies a recursive link but leaves additivity unenforced.

The contribution of this paper is to provide an explicit intertemporal accounting operator that satisfies all four requirements simultaneously. Section 4 introduces such a recursive-additive closure and shows how it reconciles superlative growth measurement with exact level additivity.

4 Recursive Aggregation Operator

4.1 Definition of the Recursive Aggregation Operator

This section constructs an explicit intertemporal accounting operator that resolves the mapping problem identified in Section 3 and satisfies the requirements set out in Section 3.3. The objective is not to propose an alternative index-number formula, but to supply a missing accounting closure that maps accepted growth-rate primitives into a globally coherent and exactly additive level system.

The key design principle is simple but consequential. Exact additivity is enforced at the level of quantities by construction, while aggregate growth is treated as an implied object of the closed system rather than as an additional primitive. This separation mirrors the distinction between local measurement and global accounting emphasized in Section 3.

Let $\{g_{i,t}\}_{i=1}^{N}$ denote the observed component log-growth rates between $t-1$ and t, constructed using a superlative index formula and treated as primitives. Let $Q_{i,t}$ denote the constructed real level of component i at time t. The aggregate level is defined endogenously through the adding-up identity $Q_t \equiv \sum_{i=1}^{N} Q_{i,t}$.

Definition 1 (Recursive-Additive Accounting Operator).
Fix a normalization date $t = 0$ and an initial level vector $\{Q_{i,0}\}_{i=1}^{N}$ satisfying $Q_0 = \sum_i Q_{i,0}$. Given the component growth primitives $\{g_{i,t}\}$, define the component levels recursively for all $t \geq 1$ by

$$Q_{i,t} \;=\; Q_{i,t-1}\exp(g_{i,t}), \qquad i = 1, \ldots, N, \tag{3}$$

and define the aggregate level at each date by

$$Q_t \;\equiv\; \sum_{i=1}^{N} Q_{i,t}. \tag{4}$$

The implied aggregate log-growth rate of the closed additive system is then

$$\tilde{g}_t \;\equiv\; \log\left(\frac{Q_t}{Q_{t-1}}\right). \tag{5}$$

Definition 1 makes clear what is taken as primitive and what is implied: component superlative growth rates are preserved exactly by (3), while aggregate growth is endogenously determined by the adding-up closure (4)–(5).

4.2 Exact Additivity and Intertemporal Closure

By construction, the operator enforces exact additivity at every date: (4) holds identically for all t, satisfying Requirement R1. Moreover, (3) provides an explicit accumulation law that maps growth primitives into levels, yielding a closed intertemporal system and satisfying Requirement R3.

This closure differs fundamentally from existing additive decompositions of superlative indexes. Contribution-based methods impose adding-up only locally for growth rates or within a benchmark comparison. By contrast, the recursive-additive operator embeds additivity directly in the intertemporal law of motion for levels. Additivity is therefore global rather than local, and holds mechanically at every date without reliance on linearization or benchmark-specific conventions.

4.3 Aggregate Growth Fidelity and the Source of Discrepancy

The remaining issue concerns aggregate growth. Because the aggregate level is defined bottom-up as a sum, the implied aggregate growth $\tilde{g}_t$ need not coincide exactly with the published chain-weighted aggregate growth rate g_t except under special conditions. This divergence should not be interpreted as a failure of the operator. Rather, it reflects the resolution of an otherwise under-specified system: one cannot generally preserve both the published aggregate growth path and exact component additivity without an explicit closure. By enforcing additivity and allowing aggregate growth to adjust endogenously, the recursive-additive operator makes this trade-off explicit.

A useful representation follows from rewriting (5). Define RA weights

$$w_{i,t-1} \;\equiv\; \frac{Q_{i,t-1}}{\sum_{j=1}^{N} Q_{j,t-1}}, \qquad \sum_{i=1}^{N} w_{i,t-1} = 1. \tag{6}$$

Then

$$\tilde{g}_t \;=\; \log\left(\sum_{i=1}^{N} w_{i,t-1}\exp(g_{i,t})\right). \tag{7}$$

Expression (7) shows that the operator aggregates component growth through a log-sum-exp mapping induced by additive levels. The published superlative aggregate growth rate g_t instead arises from a two-period superlative formula with time-varying value-share weights, so an exact match is not generally expected in multi-period level systems (Pollak 1975; Balk and Reich 2008).

4.4 Relation to Superlative Index Numbers: Second-Order Fidelity

The key point is that any discrepancy between the recursive-additive (RA) aggregate growth rate and the published superlative aggregate growth rate is structured and quantitatively small. Superlative index theory establishes that superlative growth rates provide a second-order accurate approximation to the log change of an unknown, linearly homogeneous aggregator under standard regularity conditions (Diewert 1978; Caves et al. 1982). This result is inherently local and applies to two-period growth comparisons.

The RA operator preserves component-level superlative growth primitives exactly and aggregates them through the log-sum-exp mapping in (7). Under smooth share dynamics, the discrepancy between the implied RA aggregate growth rate and the published superlative aggregate growth rate arises only from higher-order terms associated with time variation in value shares.

Proposition 1 (Second-Order Aggregate Fidelity).

Assume that (i) component growth rates $\{g_{i,t}\}_{i=1}^{N}$ are generated by a superlative index formula applied to a twice continuously differentiable, linearly homogeneous aggregator, and (ii) the associated value-share vectors evolve smoothly so that adjacent-period changes Δs_t are bounded. Then the aggregate growth rate implied by the recursive-additive system satisfies

$$\tilde{g}_t \;=\; g_t \;+\; O\big(\|\Delta s_t\|^2\big), \tag{8}$$

where Δs_t denotes the change in the relevant value-share vector between periods $t-1$ and t.

Proposition 1 establishes that the recursive-additive closure preserves the economic content of superlative aggregation to the strongest extent possible in a multi-period additive system. Exact equality with the published superlative aggregate growth rate is neither required nor theoretically justified in a globally additive framework. What superlative theory guarantees is second-order accuracy under smooth share dynamics, and this is precisely the level of fidelity delivered by the RA operator.

Proof

Let $G(Q_{1,t}, \ldots, Q_{N,t})$ denote the unknown aggregator function whose log change is locally approximated by the superlative index. By the Quadratic Approximation Lemma (Diewert 1978), the published superlative aggregate growth rate g_t provides a second-order Taylor approximation to $\Delta \log G_t$ around the midpoint share vector.

Under the recursive-additive system, the implied aggregate growth rate can be written as

$$\tilde{g}_t = \log\left(\sum_{i=1}^{N} w_{i,t-1} \exp(g_{i,t})\right),$$

where $w_{i,t-1}$ are endogenous additive weights based on lagged component levels. A first-order expansion of this expression around the superlative share vector coincides with the Divisia (and hence superlative) aggregation of component growth rates. Differences between $w_{i,t-1}$ and the superlative weighting scheme arise only through changes in shares across periods and therefore enter the expansion at second order.

Consequently, the difference between $\tilde{g}_t$ and the published superlative aggregate growth rate g_t is of order $O(\|\Delta s_t\|^2)$, establishing (8). $\square$

4.5 Normalization and Base-Year Invariance

Finally, the recursive-additive operator is invariant to normalization. Any alternative initialization $\{Q'_{i,0}\}$ with $Q'_{i,0} = c\,Q_{i,0}$ for some constant $c > 0$ generates a proportional level path $Q'_{i,t} = c\,Q_{i,t}$ and $Q'_t = c\,Q_t$ for all t. Hence, implied growth rates $\tilde{g}_t$ and share-based objects are invariant to arbitrary base-year scaling, satisfying Requirement R4.

This invariance property underlies the empirical robustness results in Section 7, where growth distortions and cumulative drift are shown to be stable across all admissible base-year normalizations.

The next section interprets these properties economically and clarifies how the recursive-additive closure reshapes the interpretation of non-additivity, growth gaps, and compositional change in chained national accounts data.

5 Properties and Economic Interpretation

Section 4 defined a recursive-additive (RA) closure that maps given component growth primitives into an exactly additive level system. The central implication is that additivity is enforced in levels by construction, while aggregate growth is an implied object of the closed system. This section derives the main interpretive consequences of that closure. In particular, we (i) show how exact additivity follows mechanically from the operator, (ii) clarify why aggregate-growth deviations are structured and typically small (second-order in share variation), and (iii) reinterpret chain non-additivity as a missing intertemporal closure rather than a failure of superlative index theory.

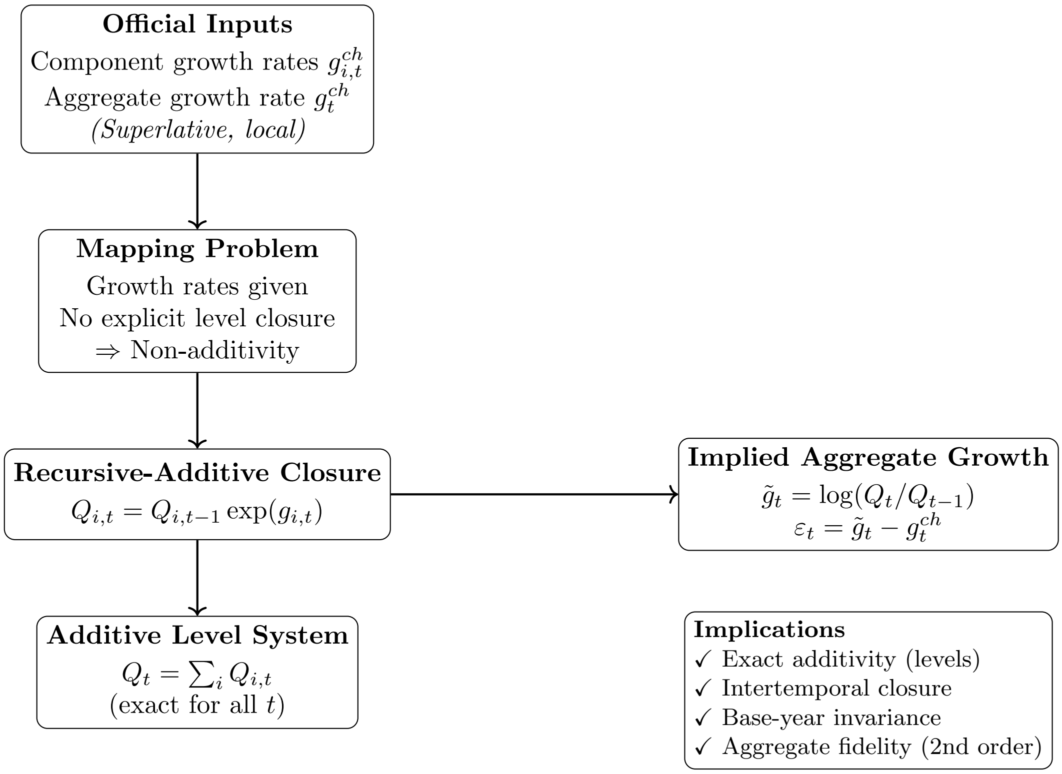


Fig. 1 RA closure: from superlative growth primitives to an exactly additive intertemporal level system.

Figure 1 summarizes the role of the RA closure: official (superlative) growth rates are treated as primitives, while an explicit level-accumulation rule supplies the missing adding-up constraint across components and time.

5.1 Exact Additivity Is Enforced by the Level Closure

The key mechanism is simple: additivity is imposed at the level law itself. Because the aggregate is defined bottom-up as $Q_t \equiv \sum_i Q_{i,t}$ at every date, exact additivity holds globally and does not depend on benchmark years, local linearizations, or within-period decompositions.

This distinguishes the RA system from additive contribution methods for superlative indexes, which typically recover adding-up only for growth contributions in a given period (or around a reference year) (Reinsdorf et al. 2002; Dumagan 2011a). In such approaches, additivity is a property of an auxiliary decomposition, not of the underlying multi-period level system. Once growth is accumulated over time, the adding-up property generally fails.

From an accounting perspective, the RA closure restores an identity that is indispensable in applications requiring level consistency—including dynamic growth accounting, multi-period sectoral decompositions, and constant-price systems where aggregation constraints must hold mechanically (Hulten 1973; Parker and Triplett 1996).

5.2 Superlative Meaning Is Preserved Through Aggregate Fidelity

A natural concern is that imposing additivity might compromise the economic meaning of superlative measures. The RA construction does not alter the component growth primitives; it preserves them exactly through the recursion. The only object that can differ from published chain accounts is the aggregate growth rate implied by the additive closure.

Section 4 showed that the implied aggregate growth can be written as a log-sum-exp aggregation of component growth with endogenous RA weights. This representation makes clear why deviations from published superlative aggregate growth are not arbitrary: they arise because two different objects are being held fixed (official aggregate growth versus exact component additivity). Under smooth share dynamics, the difference is of higher order in share variation, so first-order (and second-order) economic interpretations that motivate superlative indexes remain essentially intact (Diewert 1978; Caves et al. 1982; Balk 2004).

The interpretation is therefore not that the RA system "improves" on superlative theory, but that it embeds superlative local information in a globally consistent intertemporal accounting system.

5.3 Reinterpreting Chain Non-Additivity: A Missing Closure, Not a Contradiction

In chain-weighted national accounts, non-additivity is typically presented as an unavoidable trade-off: time-varying weights deliver economically meaningful growth rates, but chained real levels cannot be summed across components (Whelan 2000; Reinsdorf et al. 2002). The RA perspective sharpens what is missing in that description.

The standard chained system provides (i) component and aggregate growth rates and (ii) a conventional chaining rule for published aggregate levels, but it does not provide a single closed operator that simultaneously (a) preserves component primitives in levels and (b) enforces adding-up in levels at every date. In this sense, "non-additivity" is the observable symptom of an under-specified mapping from growth rates to additive level paths.

Seen this way, chain non-additivity is not an intrinsic failure of superlative index numbers (Pollak 1975; Balk and Reich 2008). It is the consequence of leaving the intertemporal level-closure problem implicit. The RA operator makes that closure explicit and thus makes the trade-off transparent: one can preserve official aggregate growth by construction, or preserve exact component additivity by construction, but not generally both.

5.4 Why an Additive Level System Matters for Growth Accounting

The practical value of the RA system becomes most evident in growth accounting and structural decomposition. Many empirical exercises implicitly require a level identity of the form

$$Q_t = \sum_i Q_{i,t}$$

to interpret sectoral contributions, reconcile input–output style aggregates, or impose accounting constraints in dynamic models. Under chained real levels, such identities do not hold except at special reference points, complicating multi-period decomposition and potentially generating composition artifacts when relative prices shift (Hulten 1973; Boyd and Roop 2004).

The RA closure restores the accounting identity at every date, allowing sectoral real quantities and their cumulative contributions to be interpreted without ad hoc rebasing or residual "statistical discrepancy" terms. This is the precise sense in which RA supplies an intertemporal accounting infrastructure that standard chained systems do not provide.

5.5 RA as an Accounting Operator Rather than an Index Formula

Finally, it is useful to emphasize the conceptual status of the approach. The RA closure is not a new superlative index formula and does not seek to replace Fisher/Törnqvist theory. Instead, it is a dynamic accounting operator that takes accepted growth-rate primitives and delivers a level system suitable for applications that demand exact adding-up across components and time.

This framing aligns the approach with the treatment of aggregates as recursively evolving state variables in macroeconomic accounting, while remaining fully compatible with the local optimality logic that motivates superlative measurement (Diewert 1978; King and Rebelo 1989).

6 Empirical Illustration: Measuring and Correcting Non-Additivity

This section illustrates the recursive-additive (RA) closure in a standard chain-weighted national accounts setting. The empirical objective is deliberately narrow: (i) document the magnitude and persistence of non-additivity when published chained real levels are summed across components, (ii) construct an exactly additive level system using the RA recursion, and (iii) quantify the resulting deviation in implied aggregate growth relative to the published chain aggregate.

The motivation is practical. Additivity is routinely required for growth accounting, contribution analysis, and constant-price consistency, yet chained volume measures are not additive by construction (Balk and Reich 2008; Dumagan 2011b). In chained systems, deflation and aggregation generally do not commute when relative prices vary across components (Balk and Reich 2008), so the published aggregate level need not equal the sum of published component levels away from the reference year.

6.1 Data and Notation

Let $i = 1, \ldots, N$ index components (industries or expenditure categories) and $t = 0, 1, \ldots, T$ index time. For each component we observe nominal values $V_{i,t}$ and published chain-weighted real levels $Y_{i,t}^{ch}$; for the aggregate we observe the published chain-weighted real level Y_t^{ch}. Throughout, "ch" denotes the published chain-weighted (superlative) series as reported in the national accounts.

We define the published (implied) growth rates by log-differences of the published chained levels:

$$g_t^{ch} \equiv \log\left(\frac{Y_t^{ch}}{Y_{t-1}^{ch}}\right), \qquad g_{i,t}^{ch} \equiv \log\left(\frac{Y_{i,t}^{ch}}{Y_{i,t-1}^{ch}}\right). \tag{9}$$

This matches empirical practice when chained real levels are taken as primitives and growth is computed mechanically. The RA construction below takes $\{g_{i,t}^{ch}\}$ as the component growth primitives.

6.2 Measuring Non-Additivity in Published Chained Levels

We quantify non-additivity via the time-t accounting discrepancy between the published chained aggregate level and the sum of published chained component levels:

$$\Delta_t^{ch} \equiv Y_t^{ch} - \sum_{i=1}^{N} Y_{i,t}^{ch}. \tag{10}$$

Because the scale of Δ_t^{ch} depends on units and the reference normalization, we report two normalized diagnostics.

(i) Relative level discrepancy.

$$\delta_t^{ch} \equiv \frac{\Delta_t^{ch}}{Y_t^{ch}}. \tag{11}$$

This object is directly interpretable as the percent gap between the published chain-weighted aggregate level and the sum of published chain-weighted component levels.

(ii) Composition distortion.

Define two mechanically different "real shares":

$$s_{i,t}^{ch} \equiv \frac{Y_{i,t}^{ch}}{Y_t^{ch}}, \qquad \bar{s}_{i,t}^{ch} \equiv \frac{Y_{i,t}^{ch}}{\sum_j Y_{j,t}^{ch}}. \tag{12}$$

When additivity fails, $s_{i,t}^{ch} \neq \bar{s}_{i,t}^{ch}$ even though both objects are often informally interpreted as "real shares." We summarize the resulting ambiguity using an L_1

distance:

$$D_t^{ch} \equiv \frac{1}{2} \sum_{i=1}^{N} \left| s_{i,t}^{ch} - \bar{s}_{i,t}^{ch} \right| . \tag{13}$$

Note that $D_t^{ch} \in [0,1]$ and equals zero if and only if the published chained levels add up exactly. Large values indicate that measured composition depends materially on whether one normalizes by the published aggregate or by the sum of published components, a concern emphasized in practical discussions of chained dollars (Landefeld et al. 2003).

6.3 Recursive-Additive (RA) Level System

We now construct an exactly additive level system using the RA closure, taking component growth rates as primitives.

Step 1: choose an additive benchmark year.

Fix a base period $t = b$ and allocate the published aggregate level across components using weights $\omega_{i,b}$ that sum to one:

$$Y_{i,b}^{RA} \equiv \omega_{i,b} \, Y_b^{ch}, \qquad \sum_{i=1}^{N} \omega_{i,b} = 1. \tag{14}$$

In the baseline implementation, $\omega_{i,b}$ is set to the component's nominal share in year b, i.e. $\omega_{i,b} = V_{i,b} / \sum_j V_{j,b}$, but alternative initializations are examined in robustness exercises.

Step 2: accumulate component levels using component growth primitives.

For $t > b$, define

$$Y_{i,t}^{RA} = Y_{i,t-1}^{RA} \exp\!\left(g_{i,t}^{ch}\right) , \tag{15}$$

and (if needed) for $t < b$, define the backward recursion analogously.

Step 3: enforce additivity by definition.

At each date, define the aggregate as the sum of components:

$$Y_t^{RA} \equiv \sum_{i=1}^{N} Y_{i,t}^{RA}. \tag{16}$$

This delivers exact additivity for all t by construction.

6.4 Growth-Rate Fidelity Relative to the Published Chain Aggregate

Because the RA aggregate is defined bottom-up, its implied aggregate growth rate is endogenous:

$$g_t^{RA} \equiv \log\left(\frac{Y_t^{RA}}{Y_{t-1}^{RA}} \right) . \tag{17}$$

We compare this implied growth to the published chain growth g_t^{ch} using:

(i) pointwise growth gap

$$\varepsilon_t \equiv g_t^{RA} - g_t^{ch}, \tag{18}$$

and

(ii) cumulative drift in levels (log).

$$\Gamma_t \equiv \log\left(\frac{Y_t^{RA}}{Y_t^{ch}}\right). \tag{19}$$

Even if ε_t is small period-by-period, Γ_t can accumulate over long samples because Y_t^{RA} and Y_t^{ch} correspond to different intertemporal closures: the published series preserves the published aggregate growth path by construction, whereas the RA series preserves exact component additivity by construction (Reinsdorf et al. 2002; Balk and Reich 2008). Reporting both objects makes this trade-off transparent.

7 Empirical Results

This section evaluates the empirical implications of the recursive-additive (RA) aggregation framework using U.S. industry value-added data. The analysis proceeds in four steps. First, we quantify the magnitude of non-additivity in published chain-weighted levels and compare it to the growth-rate distortion induced by enforcing exact additivity. Second, we examine the time-series structure of these discrepancies. Third, we document the mechanism linking non-additivity to compositional change in nominal value shares. Fourth, we assess implications for composition analysis and the robustness of the RA closure to base-year normalization.

Table 1 Summary statistics: published non-additivity $|\delta_t^{ch}|$ versus RA growth distortion $|\varepsilon_t|$

Statistic	$\lvert\delta_t^{ch}\rvert$	$\lvert\varepsilon_t\rvert$
Mean	1.30%	0.218%
Median	0.62%	0.114%
P90	3.29%	0.573%
Max	5.30%	0.787%

Table 1 provides a compact comparison between the accounting inconsistency embedded in published chain-weighted levels and the growth adjustment required to restore exact additivity. Across all statistics, $|\delta_t^{ch}|$ is materially larger than $|\varepsilon_t|$: the 90th percentile of non-additivity exceeds three percent, whereas the 90th percentile of growth distortion remains below one percent and the maximum growth distortion remains below one percent as well. The figures below show where in time these differences arise and why.

7.1 Magnitude: Published Non-Additivity versus RA Growth Distortion

Figure 2 compares the distributions of two quantities. The first is the absolute relative non-additivity in published chain-weighted levels,

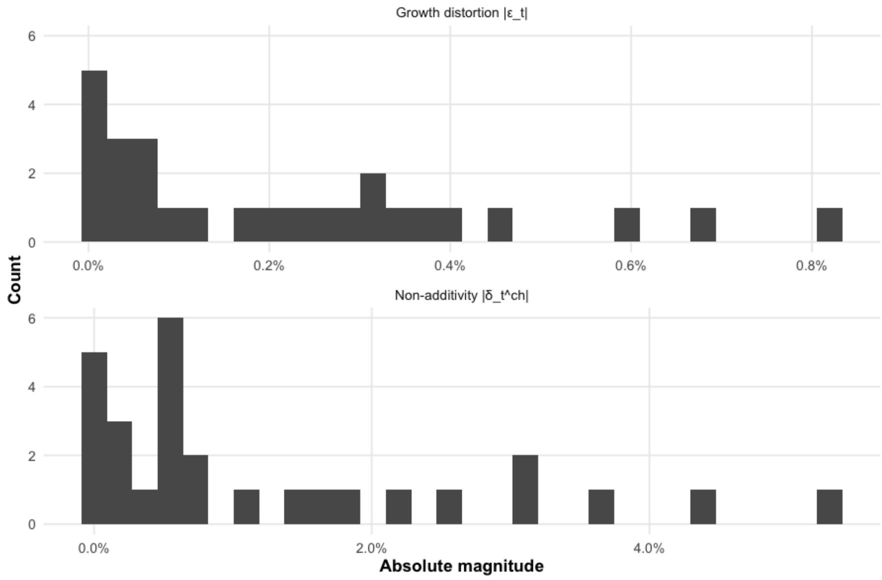
$$|\delta_t^{ch}| \equiv \left| \frac{Y_t^{ch} - \sum_i Y_{i,t}^{ch}}{Y_t^{ch}} \right|,$$

which measures the extent to which the published chained aggregate violates exact adding-up. The second is the absolute RA growth distortion,

$$|\varepsilon_t| \equiv |g_t^{RA} - g_t^{ch}|,$$

which measures how much the aggregate growth rate must adjust when exact additivity is imposed in levels.

Fig. 2 Magnitude comparison: published non-additivity versus RA growth distortion

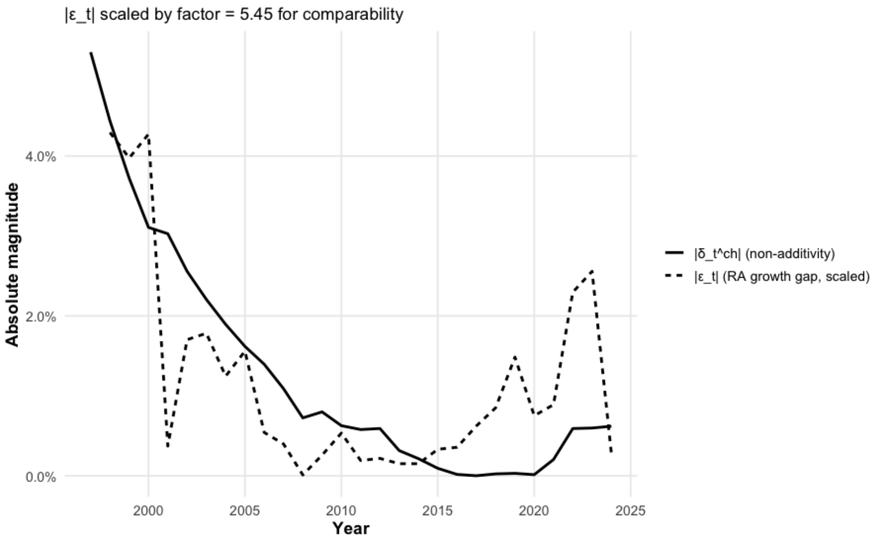


The distributions in Figure 2 differ sharply in both location and tail behavior. Published non-additivity exhibits a pronounced right tail driven by the late 1990s, when $|\delta_t^{ch}|$ reaches 5.30% (1997), 4.44% (1998), 3.72% (1999), and 3.11% (2000). By contrast, the growth distortion required to enforce exact additivity remains bounded and considerably smaller: even in those same years, $|\varepsilon_t|$ is below one percent (approximately 0.73–0.79% in 1998–2000). After 2000, $|\varepsilon_t|$ compresses further toward zero (e.g., about 0.07% in 2001 and essentially zero in 2008), whereas non-additivity remains economically nontrivial for several more years. Thus, the distributional evidence is not merely descriptive: it shows that the RA closure replaces an occasionally large level inconsistency with a comparatively small adjustment to aggregate growth.

7.2 Time Structure: Non-Additivity Is Episodic, While Growth Distortion Is Bounded

Figure 3 plots the time series of $|\delta_t^{ch}|$ and $|\varepsilon_t|$ (scaled for visual comparability). Two facts are immediate. First, non-additivity is highly episodic and persistent within episodes: it is large throughout 1997–2003 (above 2% each year) and then declines steadily toward zero through the mid-2010s, reaching 0.017% in 2016 and 0.001% in 2017. Second, non-additivity rises again in the early 2020s (0.20% in 2021 and roughly 0.59–0.60% in 2022–2023), indicating that it is not a one-off artifact of a single vintage of data.

Fig. 3 Time structure: non-additivity is episodic; RA growth distortion remains bounded



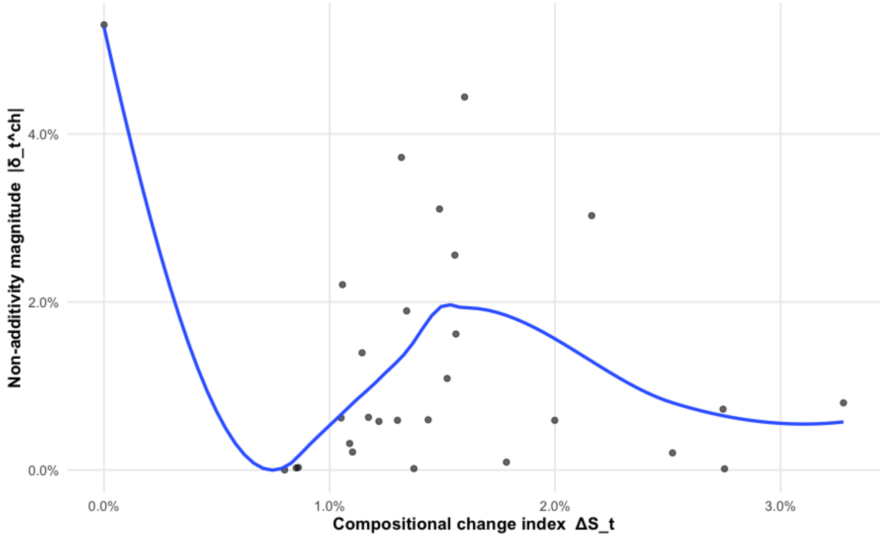
The implied growth distortion does not inherit the full amplitude of the non-additivity cycle. Even when $|\delta_t^{ch}|$ is extremely large (3–5% in 1997–2000), $|\varepsilon_t|$ remains below one percent, and in later periods it is often on the order of a few basis points (e.g., around 0.002% in 2008). This separation in time-series behavior reinforces the main empirical message: exact additivity can be restored without inducing large or unstable deviations in aggregate growth.

7.3 Mechanism: Compositional Change Drives Non-Additivity

Figure 4 relates non-additivity to the compositional change index ΔS_t (denoted DS), defined from changes in nominal value-share vectors. The relationship is strongly positive in the data. The late-1990s episode combines high compositional change—DS

around 4.0–4.3% in 1998–2000—with very large non-additivity (3–5%). When compositional change collapses in 2008 ($DS \approx 0.011\%$), non-additivity is much smaller (about 0.72%). The resurgence of non-additivity in 2022–2023 (about 0.59–0.60%) also occurs during a renewed increase in compositional change (DS roughly 2.30–2.56%).

Fig. 4 Mechanism: non-additivity increases with nominal share reallocation

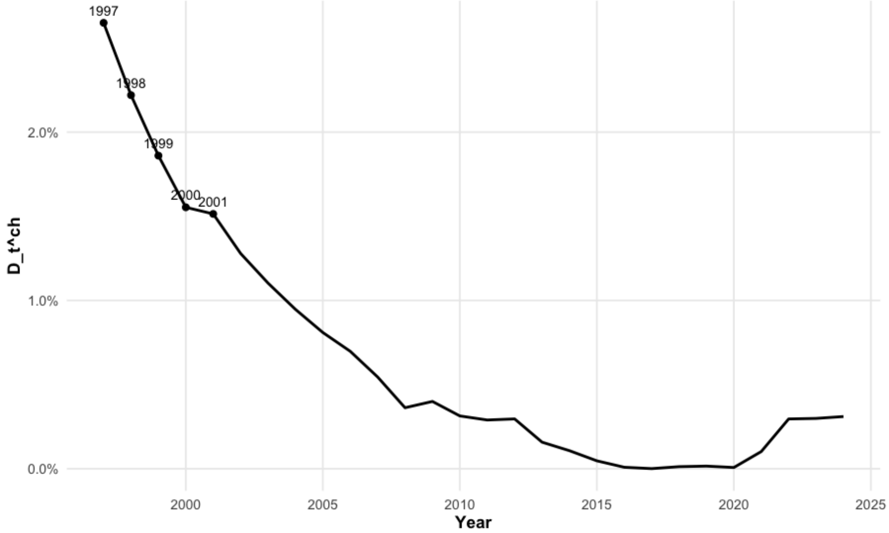


This evidence directly supports the structural interpretation of non-additivity emphasized in the index-number literature: when nominal shares and relative prices are being reallocated across components, chaining with time-varying weights generates level inconsistencies because deflation and aggregation do not commute.

7.4 Composition Distortion and the Ambiguity of Real Shares

Non-additivity has immediate consequences for composition analysis. Figure 5 plots the composition distortion index D_t^{ch}, which measures the discrepancy between "real shares" computed using the published aggregate level versus those computed using the sum of component levels. The distortion is economically meaningful in high-nonadditivity periods: it equals 2.65% in 1997 and 2.22% in 1998, declines to effectively zero by 2016–2019 (0.0086% in 2016 and 0.0005% in 2017), and rises again in the early 2020s (about 0.10% in 2021 and roughly 0.30% in 2022–2024).

Fig. 5 Composition distortion: real shares depend on the analyst's normalization



Because D_t^{ch} compares two mechanically reasonable normalizations, these magnitudes imply that in high-nonadditivity episodes even basic statements about "real industry shares" become normalization-dependent. An exactly additive level system removes this ambiguity by construction.

7.5 Cumulative Drift and Base-Year Robustness

Enforcing exact additivity implies a different intertemporal closure than the published chained aggregate. Figure 6 evaluates the resulting cumulative drift,

$$\Gamma_t \equiv \log\left(\frac{Y_t^{RA}}{Y_t^{ch}}\right),$$

across admissible base-year normalizations. The dispersion across base years is stable: the 10–90 percentile band remains bounded, while the median drift stays positive and evolves smoothly. The median drift declines from about 3.83% in 1997 to around 0.46–0.78% in 2013–2016, before increasing again to roughly 1.74% by 2024. Even at the upper tail, the 90th percentile remains below about 8% throughout the sample (e.g., 6.70% in 1997 and about 7.75–7.92% in 2022–2024).

Fig. 6 Cumulative drift (Γ_t) is stable across base-year normalizations

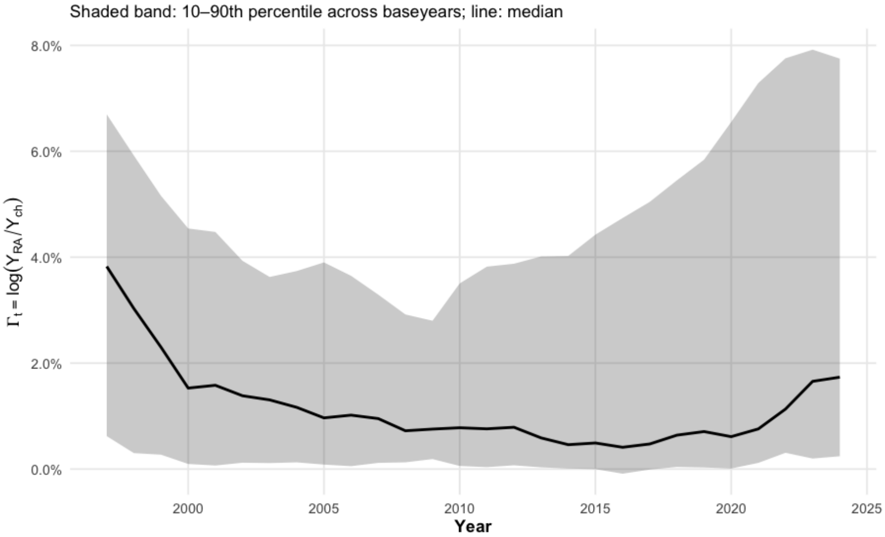
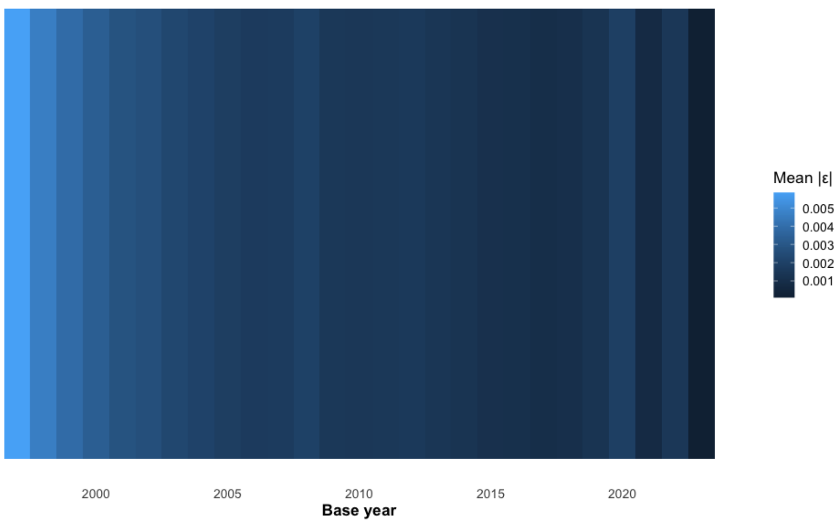


Figure 7 complements this evidence by summarizing base-year robustness in growth space. The mean absolute growth distortion $\mathbb{E}(|\varepsilon_t|)$ varies predictably with the base year: it is larger for early base years (0.58% in 1997), falls to about 0.22–0.26% for many mid-sample base years, and rises modestly for the latest base years (0.31% in 2024). Importantly, however, these magnitudes remain uniformly small relative to the non-additivity they replace.

Fig. 7 Robustness to base-year normalization: mean $|\varepsilon_t|$ across base years



7.6 Interpretation

The empirical evidence establishes five results. First, published chain-weighted levels can exhibit large and persistent non-additivity, reaching 3–5% in the late 1990s. Second, non-additivity is episodic and aligns with periods of heightened compositional change in nominal shares. Third, restoring exact additivity through the RA closure requires only small adjustments in aggregate growth, remaining below one percent even in the upper tail. Fourth, non-additivity induces a measurable ambiguity in real composition, as reflected in D_t^{ch}, which becomes economically meaningful precisely when non-additivity is large. Fifth, the implied drift and growth distortions under RA are stable across admissible base-year normalizations.

Together, these results support the paper's central claim: the apparent tension between superlative growth measurement and exact additivity reflects an intertemporal accounting closure choice rather than an empirical necessity.

8 Discussion and Practical Implications

The paper's argument can be summarized in a single distinction: superlative index theory provides locally meaningful (two-period) growth-rate primitives, but standard chaining conventions do not by themselves specify an explicit intertemporal level closure that guarantees adding-up identities. The recursive-additive (RA) operator proposed in this paper supplies that missing closure by preserving published component growth exactly while enforcing level additivity at every date.

The empirical results give this distinction immediate practical content. Published chained levels exhibit economically meaningful non-additivity (Figure 2), the deviations are episodic and persist over multi-year intervals (Figure 3), and their magnitude co-moves strongly with nominal share reallocation (Figure 4). At the same time, imposing exact additivity via the RA closure induces only small aggregate growth deviations, and these deviations remain stable across base-year normalizations (Figures 6–7). The implication is not that official chained accounts are "wrong," but that the long-standing additivity problem is best understood as a choice of intertemporal closure rather than an unavoidable empirical trade-off.

8.1 Reframing the Additivity–Superlative Trade-Off

The conventional interpretation in national accounting is that exact additivity in levels must be sacrificed in order to retain economically meaningful (superlative) growth rates (Whelan 2000; Balk and Reich 2008). The analysis here suggests a more precise diagnosis.

Superlative index theory is inherently local and two-period: it justifies adjacent-period growth comparisons but does not uniquely determine a multi-period system of real levels. Standard chained accounts implicitly close the system by privileging the published aggregate growth path; as a consequence, the sum of chained components need not equal the published chained aggregate outside the reference normalization. In this sense, "non-additivity" is not a paradox of index-number theory but the observable outcome of leaving the intertemporal mapping from growth rates to levels implicit.

Consistent with this diagnosis, Figure 4 shows that non-additivity becomes large precisely in periods of substantial nominal share reallocation, when the tension between time-varying weights and level adding-up is most acute.

The RA framework makes the closure explicit by reversing the priority: it preserves the published component growth primitives exactly and closes the system by enforcing level additivity at every date. The associated aggregate growth rate becomes an implied object of the closed additive system. Empirically, this alternative closure eliminates economically meaningful level inconsistencies at the cost of only small growth-rate deviations.

8.2 Implications for Applied Growth Accounting and Decomposition

Many applied exercises require identities of the form $Q_t = \sum_i Q_{i,t}$ to hold mechanically, including multi-period growth accounting, sectoral decomposition, constant-price input–output consistency, and dynamic accounting constraints in quantitative models. When chained levels fail to add up, practitioners often resort to ad hoc rebasing, residual discrepancy terms, or contribution-to-growth decompositions that are locally additive but do not close the accounting system in levels over time.

The RA system provides a transparent alternative accounting layer. Because additivity is enforced by construction, the decomposition of aggregate real activity into additive real components is well-defined at every date. This directly resolves the ambiguity documented by the composition distortion index D_t^{ch} (Figure 5): when published chained levels are non-additive, even the notion of a "real share" is not uniquely pinned down because the implied aggregate real level depends on the normalization convention. Under RA, shares computed from real levels are uniquely defined and consistent over time.

8.3 Interpreting the Growth Deviations: Structured and Empirically Small

A natural concern is whether enforcing additivity undermines the usefulness of superlative measurement by distorting aggregate growth. The results clarify why this concern is largely unwarranted.

The deviation between RA aggregate growth and the published chain aggregate growth reflects the difference between two closures: (i) a convention that preserves the published aggregate growth path by construction versus (ii) a convention that preserves exact component additivity by construction. Under smooth share dynamics, the discrepancy is a higher-order term in share variation, and empirically it is small: for example, the 90th percentile of published non-additivity is 3.29% while the 90th percentile of the RA growth distortion is 0.573% (Table 1), and the maximum growth distortion remains below one percent. Thus, the RA closure preserves the economic content of superlative measurement while eliminating large and practically relevant level inconsistencies.

8.4 Relation to Official Statistics and Practical Use

The RA framework is not proposed as a replacement for official chained accounts. Statistical agencies publish chained series for good reasons, including communication standards and continuity with established reporting practice. Rather, RA is best viewed as an analytical complement: it takes published component growth primitives as inputs and produces an exactly additive level system suited to applications where an explicit accounting closure is required.

Operationally, an RA implementation requires (i) published component growth rates and (ii) an initial allocation of the aggregate level across components at a benchmark date (e.g., using base-year nominal shares). The resulting series can then be used alongside the published chained series whenever exact adding-up in levels is required.

8.5 Limitations and Directions for Future Research

Two limitations deserve emphasis. First, RA requires an initial allocation of the aggregate level across components at a benchmark date. While different initializations correspond to different normalizations, the empirical fan chart shows that the implied drift and growth deviations remain stable across admissible base-year choices (Figures 6–7). Second, because RA closes the system through level additivity, the implied aggregate series need not coincide exactly with the published chained aggregate outside the benchmark; this divergence is the accounting counterpart of restoring additivity.

Future work may extend the framework to mixed-frequency settings, quarterly chain-linking, and chain-linked input–output systems, where non-additivity issues are particularly salient. Another promising direction is to examine how measurement error in components propagates differently under alternative closure conventions.

8.6 Concluding Remarks

Superlative index numbers excel at measuring economically meaningful growth rates; they do not, by themselves, define a globally additive system of real levels over time. When growth primitives are accumulated without an explicit intertemporal closure, non-additivity in levels is a predictable outcome, particularly during periods of rapid compositional change.

By supplying a recursive-additive closure, this paper shows that exact additivity and superlative economic meaning need not be viewed as mutually exclusive in applied work. The apparent tension reflects an implicit accounting choice. Making that choice explicit clarifies both the source of chained non-additivity and the options available to researchers who require coherent additive structures.

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- Author contribution: Park contributed to the conceptualization, methodology development, data curation, formal analysis, software implementation, validation, visualization, and original draft preparation.

A Proof of Proposition 1

This appendix provides a proof of Proposition 1. The goal is to formalize the sense in which the recursive-additive (RA) implied aggregate growth rate differs from the published superlative aggregate growth rate only at second order under standard smoothness and local small-change conditions.

A.1 Setup and regularity

Consider N components with prices and quantities $(p_{i,t}, q_{i,t})_{i=1}^N$ and nominal values $v_{i,t} = p_{i,t}q_{i,t}$, aggregate nominal value $V_t = \sum_i v_{i,t}$, and value shares

$$s_{i,t} \equiv \frac{v_{i,t}}{V_t}, \qquad s_t \equiv (s_{1,t}, \ldots, s_{N,t})'. \tag{20}$$

Let $\Delta x_t \equiv x_t - x_{t-1}$ and define log changes

$$\Delta \ln p_{i,t} \equiv \ln p_{i,t} - \ln p_{i,t-1}, \qquad \Delta \ln q_{i,t} \equiv \ln q_{i,t} - \ln q_{i,t-1}.$$

Write $\Delta \ln q_t \equiv (\Delta \ln q_{1,t}, \ldots, \Delta \ln q_{N,t})'$.

Assume there exists an (unknown) twice continuously differentiable, linearly homogeneous aggregator $Q(\cdot)$ such that true aggregate growth is

$$\Delta \ln Q_t \equiv \ln Q(q_t) - \ln Q(q_{t-1}).$$

Assumption A1 (Smoothness and homogeneity).

$Q(\cdot)$ is twice continuously differentiable and linearly homogeneous on a neighborhood containing $\{q_{t-1}, q_t\}$.

Assumption A2 (Local small-change regime).

There exists a scalar step size $h_t \downarrow 0$ such that

$$\|\Delta \ln q_t\| = O(h_t), \qquad \|\Delta \ln p_t\| = O(h_t), \tag{21}$$

where $\| \cdot \|$ is a fixed norm.

Assumption A3 (Superlative aggregate index / QAL).

The published aggregate superlative log-growth rate g_t satisfies the Quadratic Approximation Lemma (QAL):

$$g_t = \Delta \ln Q_t + O(h_t^2). \tag{22}$$

Assumption A4 (Component growth primitives are second-order accurate).

The reported component growth primitives $\{g_{i,t}\}$ satisfy

$$g_{i,t} = \Delta \ln q_{i,t} + O(h_t^2) \qquad \text{for each } i. \tag{23}$$

(Equivalently, the primitive component log-growth rates are second-order accurate in the local regime.)

Assumption A5 (First-order alignment of RA weights with midpoint value shares).

Let $\bar{s}_t$ denote a midpoint share vector (e.g., $\bar{s}_{i,t} \equiv \frac{1}{2}(s_{i,t-1} + s_{i,t})$). The RA weights satisfy

$$\|w_{t-1} - \bar{s}_t\| = O(h_t), \tag{24}$$

uniformly over t in the local regime.

RA aggregation.

Given positive additive component levels at $t-1$, define RA weights

$$w_{i,t-1} \equiv \frac{Q_{i,t-1}}{\sum_{j=1}^{N} Q_{j,t-1}}, \qquad \sum_i w_{i,t-1} = 1,$$

and the implied RA aggregate log-growth rate

$$\tilde{g}_t \equiv \log\Big(\sum_{i=1}^{N} w_{i,t-1} \exp(g_{i,t})\Big). \tag{25}$$

Define also $\Delta s_t \equiv s_t - s_{t-1}$.

A.2 Two lemmas

Lemma 1 (Share changes are first order).

Under Assumption A2, the value-share change satisfies

$$\|\Delta s_t\| = O(h_t). \tag{26}$$

Proof Since $s_{i,t} = \frac{p_{i,t}q_{i,t}}{\sum_j p_{j,t}q_{j,t}}$ is smooth in $(\ln p_t, \ln q_t)$ on the relevant neighborhood and Assumption A2 implies $\|(\Delta \ln p_t, \Delta \ln q_t)\| = O(h_t)$, a first-order Taylor expansion yields $\Delta s_t = J_t(\Delta \ln p_t, \Delta \ln q_t) + O(h_t^2)$ for some bounded Jacobian J_t, hence (26). □

Lemma 2 (Second-order expansion of log-sum-exp).

Let $x_t \in \mathbb{R}^N$ satisfy $\|x_t\| = O(h_t)$ and let w be a probability vector. Then

$$\log\Big(\sum_{i=1}^{N} w_i e^{x_{i,t}}\Big) = \sum_{i=1}^{N} w_i x_{i,t} + \frac{1}{2}\sum_{i=1}^{N} w_i \big(x_{i,t} - \bar{x}_t\big)^2 + O(h_t^3), \tag{27}$$

where $\bar{x}_t \equiv \sum_i w_i x_{i,t}$.

Proof Define $\phi(\lambda) \equiv \log\Big(\sum_i w_i e^{\lambda x_{i,t}}\Big)$ for $\lambda \in [0,1]$. Then $\phi'(0) = \sum_i w_i x_{i,t}$ and $\phi''(0) = \sum_i w_i (x_{i,t} - \bar{x}_t)^2$. Since $\|x_t\| = O(h_t)$ and ϕ is smooth, Taylor's theorem with remainder yields (27). □

A.3 Proof of Proposition 1

Proof of Proposition 1 We show that $\tilde{g}_t = g_t + O(\|\Delta s_t\|^2)$.

Step 1: Replace RA primitives by true log-quantity changes.
By Assumption A4, $g_{i,t} = \Delta \ln q_{i,t} + O(h_t^2)$. The map

$$F(x) \equiv \log\Big(\sum_{i=1}^{N} w_{i,t-1} e^{x_i}\Big)$$

has gradient equal to a softmax vector and therefore is 1-Lipschitz under $\|\cdot\|_\infty$ on any neighborhood. Hence replacing $g_{i,t}$ by $\Delta \ln q_{i,t}$ in (25) changes $\tilde{g}_t$ by at most $O(h_t^2)$:

$$\tilde{g}_t = \log\Big(\sum_{i=1}^{N} w_{i,t-1} \exp(\Delta \ln q_{i,t})\Big) + O(h_t^2). \tag{28}$$

Step 2: Expand RA aggregation to second order.
Apply Lemma 2 with $x_{i,t} = \Delta \ln q_{i,t}$ and $w = w_{t-1}$. Since $\|\Delta \ln q_t\| = O(h_t)$ by Assumption A2,

$$\log\Big(\sum_{i=1}^{N} w_{i,t-1} \exp(\Delta \ln q_{i,t})\Big) = \sum_{i=1}^{N} w_{i,t-1} \Delta \ln q_{i,t} + \frac{1}{2}\mathrm{Var}_{w_{t-1}}(\Delta \ln q_t) + O(h_t^3). \tag{29}$$

Combining (28) and (29) yields

$$\tilde{g}_t = \sum_{i=1}^{N} w_{i,t-1} \Delta \ln q_{i,t} + \frac{1}{2}\mathrm{Var}_{w_{t-1}}(\Delta \ln q_t) + O(h_t^2). \tag{30}$$

Step 3: Link the published superlative index to the Divisia/Törnqvist form.
Let $\bar{s}_t$ be a midpoint share vector. Define the Divisia/Törnqvist growth form

$$g_t^D \equiv \sum_{i=1}^{N} \bar{s}_{i,t} \Delta \ln q_{i,t}.$$

Under Assumption A3 (QAL), the published superlative growth satisfies

$$g_t = g_t^D + O(h_t^2). \tag{31}$$

***Step 4: Compare* (30) *with* (31).**
Subtract g_t from $\tilde{g}_t$ using (30) and (31):

$$\tilde{g}_t - g_t = \sum_{i=1}^{N} (w_{i,t-1} - \bar{s}_{i,t}) \Delta \ln q_{i,t} + \frac{1}{2}\mathrm{Var}_{w_{t-1}}(\Delta \ln q_t) + O(h_t^2). \tag{32}$$

By Assumption A5, $\|w_{t-1} - \bar{s}_t\| = O(h_t)$, and by Assumption A2, $\|\Delta \ln q_t\| = O(h_t)$, so the first term in (32) is $O(h_t^2)$. The variance term is a quadratic form in $\Delta \ln q_t$ with bounded coefficients and is therefore also $O(h_t^2)$. Hence

$$\tilde{g}_t - g_t = O(h_t^2).$$

Step 5: Express the bound in share-change notation.
By Lemma 1, $\|\Delta s_t\| = O(h_t)$, hence $h_t^2 = O(\|\Delta s_t\|^2)$. Therefore

$$\tilde{g}_t = g_t + O(\|\Delta s_t\|^2),$$

which is (8). $\square$